2. Purple sulphur bacteria. Example: *Thiospirillum* and *Chromatium.*
3. Purple non-sulphur bacteria. Example: *Rhodopseudomonas* and *Rhodospirillum*.

Test tube funnel experiment or Experiment to prove oxygen evolved during Photosynthesis

1. Place *Hydrilla* plant at the bottom of a beaker containing water.
2. Cover the plant with an inverted funnel.
3. Invert a test tube over the funnel.
4. Keep this setup in sunlight.

Note your observations (Figure 13.23).

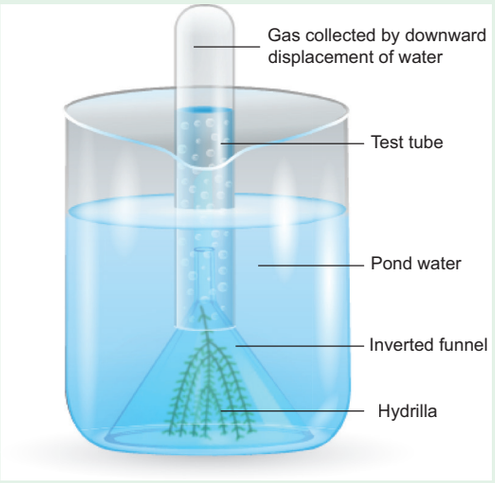


Figure 13.23: Test tube funnel experiment

Summary

Photosynthesis is an oxidation and reduction process. It has two phases: the light reaction and dark reaction. During light reaction water is oxidised to release O_2 and during dark reaction $CO_2$is reduced to form sugars. Solar energy is trapped by pigment system I and pigment system II. P700 and P680 act as reaction centres for PS I and PS II respectively. Splitting of water molecule (Photolysis) produces electrons, protons and oxygen. Photophosphorylation takes place through cyclic and non-cyclic mechanisms and generates energy and reducing power. Dark reaction or biosynthetic phase of photosynthesis use the products of light energy (ATP and NADPH + H^+) and carbon dioxide is reduced to Carbohydrates. Carbon pathway in C_3 cycle has RUBP as the acceptor molecule and the first product is PGA (3C). Carbon pathway in C_4 plants involves mesophyll and bundle sheath cells, Kranz anatomy. Dimorphic chloroplast, no photorespiration, acceptor molecule as PEP and first product as OAA (4C) are some of the unique characters of C_4 cycle. C_2 Cycle or photorespiration is operated when less amount of CO_2 is used for reduction and O_2 increases. Rubisco starts to play oxygenase role. Succulent and xerophytic plants show reverse stomatal rhythm as they open during night time and close during day time and follow CAM cycle. Night time produces malic acid and during day time malate is converted into pyruvate and produces CO_2 which is reduced to carbohydrates. Photosynthesis is affected by internal and external factors. Bacterial photosynthesis is the primitive type of photosynthesis and it involves only photosystem I.

CPZYDC

Evaluation

1. **Assertion (A):** Increase in Proton gradient inside lumen responsible for ATP synthesis
   Reason (R): Oxygen evolving complex of PS I located on thylakoid membrane facing Stroma, releases H+ ions
   a. Both Assertion and Reason are True.
   b. Assertion is True and Reason is False.
   c. Reason is True and Assertion is False.
   d. Both Assertion and Reason are False.
2. Which chlorophyll molecule does not have a phytol tail?
   a. Chl- a b. Chl-b c. Chl- c d. Chl -d
3. The correct sequence of flow of electrons in the light reaction is
   a. PS II, plastoquinone, cytochrome, PS I, ferredoxin.
   b. PS I, plastoquinone, cytochrome, PS II ferredoxin.
   c. PS II, ferredoxin, plastoquinone, cytochrome, PS I.
   d. PS II, plastoquinone, cytochrome, PS II, ferredoxin.

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